International Institute of Information Technology, Hyderabad

(Deemed to be University)

CS6.401 Software Engineering – Spring 2026

Mid Semester Examination

Max. Time: 1.5 Hr

Max. Marks: 30

Instructions to the students

1. You are **allowed** to bring up to **2 sheets (A4 size, total of 4 pages including back-to-back)** of **handwritten notes**. You are not required to submit these notes along with your answer sheets.
2. Any form of printed/scanned materials, digital notes and photocopies are not allowed.
3. Borrowing notes from other students in the examination hall is prohibited.
4. Answers written (or figures made) **using pencils won't be considered** for evaluation.
5. Please **read the descriptions** of the questions (scenarios) carefully. While answering please ensure that any **assumptions made are clearly stated**.
6. There are a total of five questions. Please note that questions one and two are MCQs (requires 2-3 lines explanation) and carries 2.5 marks each. Out of the rest three, one carries 5 marks and the remaining two questions carry 10 marks each.

Good Luck

QuickKart: A Delivery Platform

Congratulations! You have been invited as a *Design Reviewer* for *QuickKart*, a quick-commerce platform that delivers groceries and essential items within 10–15 minutes from nearby dark stores (the stores that are only accessible to the team and not common people). The system supports product browsing, cart management, order placement, inventory validation, payment, delivery partner assignment, and real-time tracking.

QuickKart now plans to expand its features and wants to review the current design to identify possible design issues and technical debt (if any). The upcoming features include support for new payment providers, flexible delivery allocation strategies, improved coupon handling, and integration of a *Generative AI-based multilingual chatbot* that can answer customer queries, assist in placing orders, and track deliveries in different Indian languages.
QuickKart has assembled a team of 15 software engineers. The team has started development, but poor design practices have led to challenges related to *maintainability, extensibility, understandability and testability.*

As a design reviewer, you are expected to analyze and improve the system design. You have 90 minutes to complete five critical tasks, with your final score **(out of 30 points)** determining your final payment.

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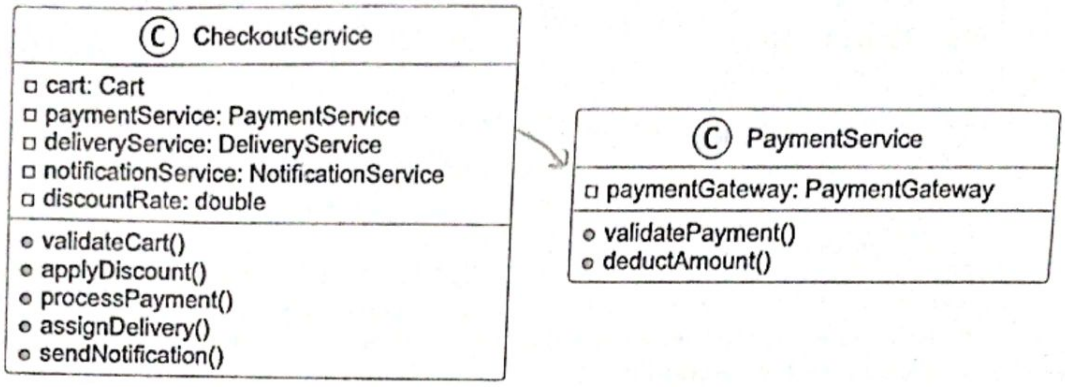
1. You started by going through their existing design available as UML diagrams. You noticed the below and you felt that there are some design smells. Which option best identifies the most relevant design smell and why ? (select the most appropriate one) **(2.5 points)**



a) Modularization Smell ✓
b) Hierarchy Smell
c) Abstraction Smell
d) Encapsulation Smell
e) None of the above

2. As part of your design review of QuickKart, you analyze the OrderService class and observe the following metrics: LOC = 860; Cyclomatic Complexity = 7; WMC (Weighted Methods per Class) = 52; CBO (Coupling Between Objects) = 21; Halstead Volume = High.

When you noticed a bit deeper you understood that the class is responsible for: i) order validation; ii) inventory reservation; iii) Coupon Application; iv) Payment Invocation; v) Delivery Partner Assignment and vi) Notification Triggering. Given the above context, Which of the following inferences are MOST appropriate and why? (You can select all that apply) **(2.5 points)**

a) Low cyclomatic complexity alone guarantees good maintainability.
b) High CBO suggests that changes in dependent services may have ripple effects.
c) High WMC indicates that the class may be handling too many responsibilities.
d) High Halstead Volume suggests increased cognitive effort to understand the implementation.
e) Since LOC is less than 1000, refactoring is not necessary.

3. QuickKart is composed of several major components including: i) Product Discovery and Catalog; ii) Cart and Checkout; iii) Order Processing and Delivery Management and iv) Payments and Refund Handling.

While the system is operational, the architecture team has realized that the *Order Processing and Delivery Management component lacks* a clear design model. This has made it difficult to reason about responsibilities, extend the system with new delivery strategies, and evaluate potential design weaknesses. You have been asked to formally

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model the *Order Processing and Delivery Management subsystem*, which is responsible for: i) Validating and confirming orders; ii) Reserving inventory from a selected dark store (these are stores that are not available for common customers but they serve as a place to store stocks and deliver to nearest customer); iii) Assigning delivery partners and iv) Updating the order status throughout the delivery lifecycle.

You are required to represent both the structural and behavioural aspects of this subsystem using appropriate UML diagrams. **(10 points)**

a. Identify **4–5 core classes** essential for this subsystem. Create a *UML Class Diagram* capturing the main classes and their relationships (associations, aggregations/compositions, generalizations, multiplicities - one-to-one, one-to-many, etc. where appropriate). (5 points)
b. Model one behavioural view (e.g. Sequence diagram or activity diagram) for any of the relevant flow such as Order confirmation, Delivery Partner Assignment or Order Status Update (2 points)
c. Highlight at least 2 design principles (GRASP or SOLID principles) that have been used in your design and briefly explain how they have been applied. (3 points)

Note: You only have to focus on *4-5 critical classes* that are most important for the given scenario. Also, provide a 2-line description of each identified class and their relationships. These classes may or may not include classes from other subsystems. Clearly state any assumptions made.

4. During your review of QuickKart, you observe the following two classes **(5 points)**:

```
public class InvoiceService
{

  public double calculateFinalAmount(double totalAmount, double discount,          double
tax)
   {
    double finalAmount = totalAmount - discount;
    finalAmount = finalAmount + tax;
    return finalAmount;
   }
}


public class RefundService
{

    public double calculateRefundAmount(double totalAmount, double discount,      double tax)
```


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```
    {
        double finalAmount = totalAmount - discount;
        finalAmount = finalAmount + tax;
        return finalAmount;
    }
}
```

a. Identify any two code smells present in this code. Explain briefly the identified smells with respect to the above code (2 points)
b. Suggest appropriate refactoring techniques to address the identified smells by providing the refactored code (2 points).
c. Explain briefly how your refactoring improves the structure of the code with emphasis on at least two quality attributes (1 points)

Note: you are free to make assumptions (but do mention them) that may drive you towards an appropriate solution. **You don't have to write the pseudocode of the entire** refactored solution; instead, use pseudocode just to highlight code-level changes.

5. As part of the design review committee, you have been asked to help with the design of *QuickAssist,* a Generative AI-based chatbot being integrated into QuickKart. QuickAssist assists users with order tracking, cancellations, refunds, and product search. The system must: i) support multiple model providers (each offering different language, costs, latency, etc in text generation); ii) Handle different provider SDK[1] formats; iii) Select which model to use at run-time based on various factors (e.g. type of query, latency of model, etc.) and iv) Allow new providers and selection mechanisms to be added in future. Your goal is to Propose a design that can evolve without frequent rewrites while maintaining other quality attributes. **(10 points)**

   Please note that, you may treat the AI model itself as a black box that takes input text and produces output text.
   a. List the key classes/interfaces you would introduce (at least 4–5) and briefly state the responsibility of each. Mention any assumptions (3 points)
   b. Describe your solution using UML diagrams or pseudocode (as and when necessary) (5 points)
   c. Briefly explain how your design improves *extensibility and understandability* (or any two quality attributes you choose), and why (2 points)

Now that you have completed the task, it's time to wait to know your final points ☺

[1] SDK (Software Development Kit): A vendor-provided client library used to integrate with their external service.

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